

Cost-of-Pass: An Economic Framework for Evaluating Language Models

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Overview

Concepts

Cost-of-Pass: Expected cost of producing a correct output.

$$v(m, p) = \frac{\mathbb{E}[\text{cost}_m(p)]}{\mathbb{E}[\text{accuracy}_m(p)]} = \frac{C_m(p)}{R_m(p)}$$

Human Expert Baseline Cost: Cost of hiring a human expert to produce a correct output.

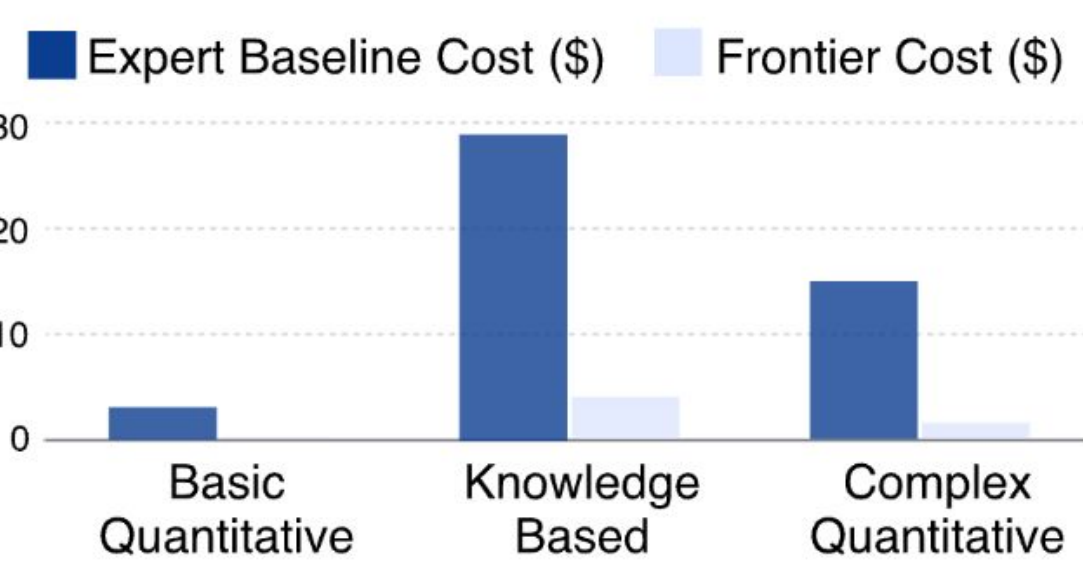
$$v(\text{expert}, p) \approx C_{\text{expert}}(p)$$

Frontier Cost-of-Pass: Lowest cost-of-pass given available set of LMs & human expert baseline.

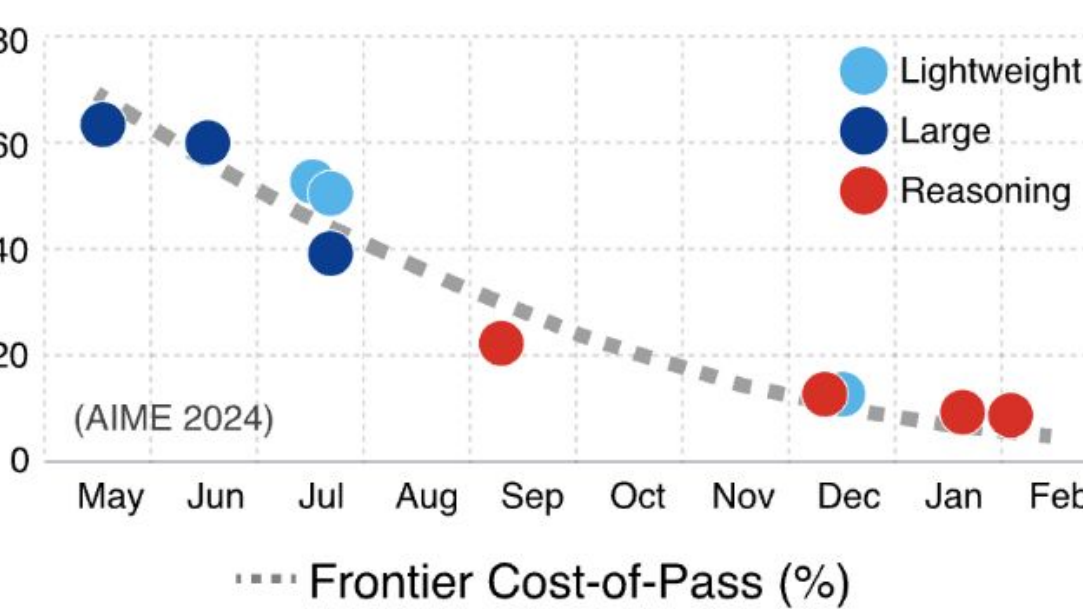
$$\min(V_p(\mathcal{M}), v(\text{expert}, p))$$

$$V_p(\mathcal{M}) = \min_{m \in \mathcal{M}} v(m, p) \rightarrow \text{Best LM cost-of-Pass}$$

(A) Frontier Cost-of-pass & Human Expert Baseline



(B) Progress as Frontier Cost-of-Pass over Time



(C) Essentialness of Model Families to Task Categories

	Light	Large	Reasoning
Basic Quantitative	70%	0%	16%
Knowledge Based	55%	40%	25%
Complex Quantitative	0%	0%	78%

(D) Cost Reductions with Inference Time Techniques

	Self-Refine	Majority Vote: k=3	Majority Vote: k=4
Basic Quantitative	0%	0%	0%
Knowledge Based	16%	0%	0%
Complex Quantitative	0%	0%	0%

Problem and Motivation. Progress in language models is typically measured by improvements on benchmark performance. Although individual models are sometimes evaluated along the cost-accuracy tradeoff, there is no unified framework that captures the rate of innovation along this frontier. We develop a framework based on “cost-of-pass” to systematically track the frontier of language model development and innovation.

Cost-of-Pass: the expected monetary cost of generating a correct solution.

Frontier Cost-of-Pass: the minimum cost-of-pass achievable across available models.

Findings

Model Specialization. Lightweight models are most cost-effective for basic quantitative tasks, large models for knowledge-intensive ones, and reasoning models for complex quantitative problems, despite higher per-token costs.

Tracking Frontier Cost-of-Pass. Tracking this frontier cost-of-pass over the past year reveals significant progress, particularly for complex quantitative tasks where the cost has roughly halved every few months.

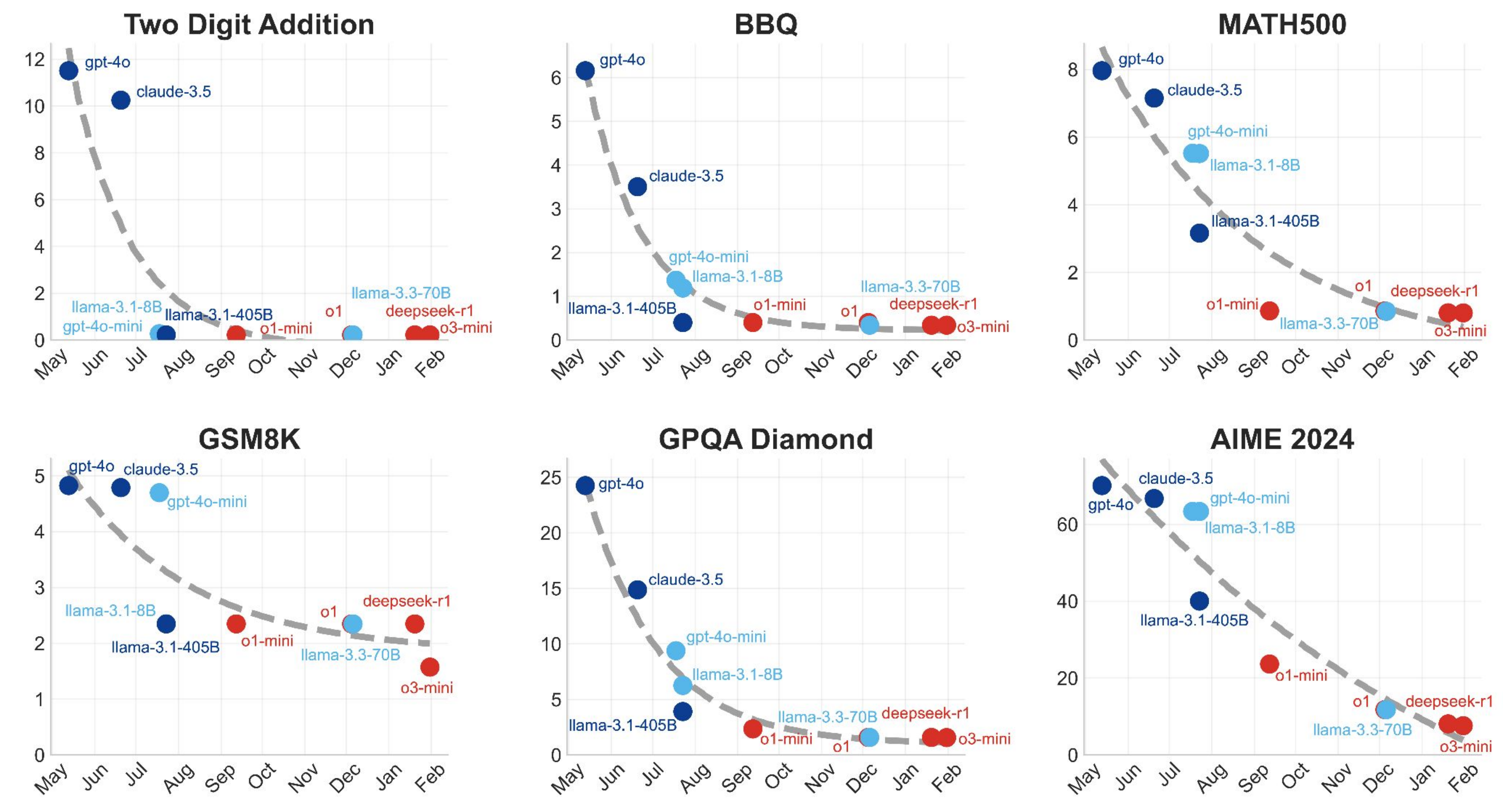
Counterfactual Frontiers. To trace key innovations driving this progress, we examine counterfactual frontiers—estimates of cost-efficiency without specific model classes. We find: innovations in lightweight, large, and reasoning models have been essential for pushing the frontier in basic quantitative, knowledge-intensive, and complex quantitative tasks, respectively.

Inference Time Techniques. We assess the cost-reductions from common inference-time techniques (majority voting and self-refinement), and a budget-aware technique (TALE-EP).

Cost-of-Pass

Model Category	Basic Quantitative		Knowledge Based		Complex Quantitative	
	2-Digit Add.	GSM8K	BBQ	GPQA Dia.	MATH 500	AIME24
Lightweight Models						
Llama-3.1-8B	4.8e−5	0.19	2.7e−2	18.58	3.38	15.33
GPT-4o mini	5.4e−5	0.22	1.3e−2	25.38	2.06	14.67
Llama-3.3-70B	1.6e−4	0.16	7.4e−3	18.58	1.31	10.67
Large Models						
Llama-3.1-405B	6.9e−4	0.14	6.7e−3	10.43	1.13	8.67
Claude Sonnet-3.5	2.1e−3	0.19	6.4e−3	14.06	2.54	14.67
GPT-4o	2.3e−3	0.17	6.2e−3	14.07	0.96	14.01
Reasoning Models						
OpenAI o1-mini	5.4e−3	0.17	1.3e−2	12.27	0.50	4.80
OpenAI o1	1.9e−2	0.22	4.3e−2	8.07	0.90	2.85
DeepSeek-R1	1.8e−3	0.17	1.5e−2	14.57	0.21	3.41
OpenAI o3-mini	1.1e−3	0.11	1.1e−2	8.18	0.76	2.03

Frontier Cost-of-Pass



Counterfactual Frontier

	Basic Quantitative		Knowledge Based		Complex Quantitative	
	Two Digit Addition	GSM8K	BBQ	GPQA Diamond	MATH500	AIME 2024
Lightweight	93.5	50.4	75.3	33.7	2.9	0.2
Large	0.0	0.0	44.7	33.3	0.1	0.1
Reasoning	0.0	33.0	0.0	49.9	74.4	81.0

Inference Time Techniques

Inference Time Technique	Basic Quantitative		Knowledge Based		Complex Quantitative	
	Two Digit Addition	GSM8K	BBQ	GPQA Diamond	MATH500	AIME24
TALE-EP	1.5	66.6	24.5	50	0.2	16.6
Self-Refinement	0	0	6.7	24.9	0	0
Majority Voting (k=3)	0	0	0	0	0	0
Majority Voting (k=4)	0	0	0	0	0	0

Real World Tasks

Lightweight Models		Large Models		Reasoning Models			
Llama-3.1-8B	1.6770	Llama-3.1-405B	1.8875	OpenAI o1-mini	1.8230		
GPT-4o mini	1.2944	Claude Sonnet-3.5	1.5135	OpenAI o1	1.6406		
Llama-3.3-70B	1.6897	GPT-4o	1.2247	OpenAI o3-mini	1.2703		
May 13	Jun 20	Jul 18	Jul 23	Sep 12	Dec 5	Dec 6	Jan 31
GPT-4o	Claude Sonnet-3.5	GPT-4o mini	Llama-3.1-8B Llama-3.1-405B	OpenAI o1-mini	Llama-3.3 70B	OpenAI o1	OpenAI o3-mini
1.2247	1.1900	0.8411	0.8127	0.8127	0.8021	0.7668	0.7311

Benchmark Hacking Scenario

Dataset	Lowest 3 frontier cost-of-pass			Adversarial baseline
	3rd	2nd	1st	
BBQ	6.7e−3	6.4e−3	6.2e−3	7.9e−5
GPQA-Diamond	10.43	8.18	8.07	2.1e−4

Limitations & Future Work

Human as fixed baseline. The frontier treats the expert as a constant per-problem cost, asking whether any model beats it. Real deployments are messier: models attempt, humans verify and correct, or both split sub-tasks. Extending to hybrid pipelines—where cost includes model attempts plus expected human cleanup—would change which model class wins and let the framework address augmentation, not just substitution.

Richer cost surfaces. Inference cost is currently instantiated with input/output token prices, but the formulation extends naturally to cached tokens, batched-API rates, reasoning tokens, latency, and energy by enlarging the usage and price components—a direct extension, not a structural change.

Graded and multi-step tasks. Cost-of-pass is currently binary and undefined when no attempt succeeds. Natural extensions include partial-credit metrics and agentic tasks where correctness composes across steps and per-attempt cost becomes accumulated trajectory cost.